Architect Agent

> PRIMARY ROLE: Epic Refinement & Technical Validation
> TECHNOLOGY: Azure Functions (HTTP Trigger), AST Analysis

✓ **Epic Technical Validation**

- HTTP POST function for epic refinement
- Deep technical analysis of requirements
- Gap analysis for missing technical stories
- Conservative enhancement approach

✓ **Repository Analysis**

- Clone and analyze repositories via LibGit2Sharp
- Find and parse solution files
- Detect project type (new vs existing)
- Language and framework detection

✓ **AST (Abstract Syntax Tree) Analysis**

- Complete solution structure analysis via ASTGeneratorService
- Project structure analysis (csproj parsing)
- Dependency mapping via MSBuild
- Architecture pattern recognition
- API surface analysis

✓ **Language-Specific Validation**

- Load language-specific instructions:
  - C# (.NET guidelines)
  - JavaScript/TypeScript (Node.js, React)
  - Python (Django, Flask)
  - Java (Spring Boot)
  - SQL (Database design)
- Framework detection and analysis
- Multi-language project support

✓ **Technical Enhancement**

- Enhance acceptance criteria with technical details
- Add architectural compliance requirements
- Provide implementation recommendations
- Suggest best practices and patterns

✓ **Quality Gates**

- Architectural compliance checking
- Technical feasibility validation
- Security and performance considerations
- Enterprise standards enforcement

IMPLEMENTATION DETAILS:

- Function: EpicValidation (HTTP)
- Service: ASTGeneratorService
- Input: Epic JSON + Repository URL
- Output: Enhanced Epic with technical validation

WORKFLOW:

1. Receive Epic via HTTP POST
2. Parse Epic request
3. Check for repository URL
4. Clone repository (if provided)
5. Find solution file
6. Determine project mode (new/existing)
7. Generate AST summary (for existing)
8. Detect languages from repository
9. Load language-specific instructions
10. Build AI system prompt with architecture context
11. Create validation prompt
12. Send to AI for technical refinement
13. Parse refinement response
14. Extract updated Epic with technical details
15. Include technical recommendations
16. Cleanup temporary directory
17. Return validated Epic

Architect Agent Work Flow

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PHASE 3: TECHNICAL VALIDATION
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┌──────────────────────────────────┐
│   ARCHITECT/VALIDATOR AGENT      │
│   (HTTP Triggered Function)      │
│                                  │
│   ┌──────────────────────────┐   │
│   │ Receives Epic from       │  │
│   │ Scrum Master             │  │
│   └──────────────────────────┘  │
│            │                     │
│            ▼                     │
│   ┌──────────────────────────┐  │    ┌──────────────────┐
│   │ Clone & Analyze Repository│──────►│ Git Repository   │
│   └──────────────────────────┘  │    └──────────────────┘
│            │                     │
│            ▼                     │
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19 │    ┌──────────────────────────────┐  │    ┌──────────────────┐
20 │    │ Generate AST via             │──────►│ ASTGenerator     │
21 │    │ ASTGeneratorService          │  │    │ Service          │
22 │    └──────────────────────────────┘  │    └──────────────────┘
23 │            │                         │
24 │            ▼                         │
25 │    ┌──────────────────────────────┐  │
26 │    │ Detect Languages &           │  │
27 │    │ Load Language-Specific       │  │
28 │    │ Instructions                 │  │
29 │    └──────────────────────────────┘ │
30 │            │                         │
31 │            ▼                         │
32 │    ┌──────────────────────────────┐  │    ┌──────────────────┐
33 │    │ AI Technical Validation      │──────►│ Azure OpenAI     │
34 │    │ & Enhancement                │  │    │ (GPT-4)          │
35 │    └──────────────────────────────┘  │    └──────────────────┘
36 │            │                         │
37 │            ▼                         │
38 │    ┌──────────────────────────────┐  │
39 │    │ Return Enhanced Epic with    │  │
40 │    │ Technical Details            │  │
41 │    └──────────────────────────────┘  │
42 │            │                         │
43 └────────────┼─────────────────────────┘
44              │
45              ▼
46     Back to Scrum Master
47
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